

#In this code, we demonstrate the existence of the curves γ_2 ($r_2=0$) and $\tilde{B}=0$ in region A5 by Lemma 6.13.

$$A := -(w+z)^3 \left((x^2 y^2 + z^4) w^4 - x^2 y^2 z^4 \right) :$$

$$B := x^2 y^5 w^2 z^2 (w+z)^2 (w-z) - x^2 y^2 (w+z)^3 (w^2 + z^2) w^2 z^2 (w-z) + w^4 y^3 z^4 (w+z)^3 + w^4 z^4 x^3 y^3 :$$

$$R := B^2 \cdot (w+z)^2 - y^2)^3 - A^2 \cdot (y^2 \cdot z \cdot w)^3$$

$$R := (x^2 y^5 w^2 z^2 (w+z)^2 (w-z) - x^2 y^2 (w+z)^3 (w^2 + z^2) w^2 z^2 (w-z) + w^4 y^3 z^4 (w+z)^3 + w^4 z^4 x^3 y^3)^2 ((w+z)^2 - y^2)^3 - (w+z)^6 ((x^2 y^2 + z^4) w^4 - x^2 y^2 z^4)^2 y^6 z^3 w^3 \quad (1)$$

#Apply the rational parametrization constructed in this section.

$$\text{parametrization} := \left\{ w = a + \frac{1}{a}, \right.$$

$$z = b + \frac{1}{b},$$

$$x = a - \frac{1}{a} + b - \frac{1}{b},$$

$$y = a - \frac{1}{a} - b + \frac{1}{b} \Big\} :$$

simplify(collect(numer(expand(subs(parametrization, R))), a));

$$\begin{aligned} & - (a^2 + 1)^3 (a - b)^6 (ab + 1)^{12} \left(b^{16} + a^{32} b^{16} - 1024 b^9 (b^2 + 1)^4 a^7 + 8 \left(b^{20} + b^{18} - 27 b^{16} \right. \right. \\ & \quad \left. \left. + 33 b^{14} - \frac{2189}{2} b^{12} - 3875 b^{10} - \frac{9357}{2} b^8 - 3231 b^6 - 2747 b^4 - 1055 b^2 - 127 \right) b^6 a^{10} \right. \\ & \quad \left. + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} - 1372 b^{18} - 9036 b^{16} - 25400 b^{14} - 54798 b^{12} - 64184 b^{10} \right. \\ & \quad \left. - 27212 b^8 - 6620 b^6 - 2636 b^4 + 36 b^2 + 1) a^{12} - 4 b^2 (b^{28} + 12 b^{26} + 37 b^{24} + 378 b^{22} \right. \\ & \quad \left. + 4187 b^{20} + 19656 b^{18} + 41514 b^{16} + 57358 b^{14} + 65514 b^{12} + 34120 b^{10} + 4315 b^8 \right. \\ & \quad \left. + 1018 b^6 - 155 b^4 + 12 b^2 + 1) a^{14} - 4 b^2 (b^{28} + 76 b^{26} - 539 b^{24} - 2118 b^{22} + 5211 b^{20} \right. \\ & \quad \left. + 36360 b^{18} + 92394 b^{16} + 78414 b^{14} + 38058 b^{12} + 17992 b^{10} - 2085 b^8 - 518 b^6 + 37 b^4 \right. \\ & \quad \left. + 12 b^2 + 1) a^{18} + 8 \left(b^{20} + b^{18} - 699 b^{16} + 2369 b^{14} - \frac{1037}{2} b^{12} - 1827 b^{10} + \frac{2163}{2} b^8 \right. \right. \end{aligned} \quad (2)$$

$$\begin{aligned}
& -159 b^6 - 27 b^4 + b^2 + 1 \Big) b^6 a^{22} - 9 \left(b^{16} + \frac{4168}{9} b^{14} - 528 b^{12} - 328 b^{10} + \frac{3698}{9} b^8 \right. \\
& \left. - 72 b^6 - 16 b^4 + 8 b^2 + 1 \right) b^8 a^{24} + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^{30} + (6 b^{20} - 32 b^{18} \\
& + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} + (-1024 b^{22} + 72 b^{20} + 692 b^{18} + 8 b^{16} - 76 b^{14} \\
& + 72 b^{12}) a^{26} + (-254 b^{28} + 840 b^{26} + 6504 b^{24} - 696 b^{22} - 25752 b^{20} - 169584 b^{18} \\
& - 181788 b^{16} - 85104 b^{14} - 50328 b^{12} - 4792 b^{10} + 360 b^8 + 72 b^6 + 2 b^4) a^{20} + (b^{32} \\
& + 8 b^{30} + 20 b^{28} + 1672 b^{26} + 1950 b^{24} - 32376 b^{22} - 157780 b^{20} - 317080 b^{18} - 279672 b^{16} \\
& - 222104 b^{14} - 125524 b^{12} - 7288 b^{10} - 3426 b^8 - 120 b^6 + 20 b^4 + 8 b^2 + 1) a^{16} + (\\
& -9 b^{24} - 72 b^{22} + 144 b^{20} - 1912 b^{18} + 142 b^{16} + 4232 b^{14} + 2192 b^{12} - 1864 b^{10} - 265 b^8) \\
& a^8 + (72 b^{20} - 76 b^{18} + 1288 b^{16} + 692 b^{14} - 696 b^{12} - 512 b^{10}) a^6 + (6 b^{20} - 32 b^{18} \\
& + 44 b^{16} - 288 b^{14} - 250 b^{12}) a^4 + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^2 - 1024 (b^2 - 5) \left(b^2 \right. \\
& \left. - \frac{1}{2} \right) (b^2 + 1)^4 b^{13} a^{23} + 2560 \left(b^6 - \frac{6}{5} b^4 + \frac{8}{5} b^2 - 1 \right) (b^2 + 1)^4 b^7 a^{11} + 512 b^7 (b^2 \\
& - 2) (b^2 + 1)^5 a^9 + 512 b^7 (b^{10} - 10 b^8 - 9 b^6 + 42 b^4 - 19 b^2 + 1) (b^2 + 1)^4 a^{15} \\
& - 2560 (b^2 + 1)^4 \left(b^8 - \frac{16}{5} b^6 + b^4 - \frac{3}{5} b^2 + \frac{3}{5} \right) b^7 a^{13} + 512 b^7 (b^{10} + 11 b^8 - 78 b^6 \\
& + 81 b^4 - 22 b^2 + 1) (b^2 + 1)^4 a^{17} - 2560 \left(b^6 - \frac{38}{5} b^4 + \frac{42}{5} b^2 - \frac{11}{5} \right) (b^2 \\
& + 1)^4 b^{11} a^{21} - 1536 \left(b^8 - \frac{43}{3} b^6 + \frac{95}{3} b^4 - \frac{52}{3} b^2 + \frac{7}{3} \right) (b^2 + 1)^4 b^9 a^{19} \Big) (a \\
& + b)^6 (b^2 + 1)^3
\end{aligned}$$

#Define r2

$$\begin{aligned}
r2 := & \left(b^{16} + a^{32} b^{16} - 1024 b^9 (b^2 + 1)^4 a^7 + 8 \left(b^{20} + b^{18} - 27 b^{16} + 33 b^{14} - \frac{2189}{2} b^{12} \right. \right. \\
& \left. \left. - 3875 b^{10} - \frac{9357}{2} b^8 - 3231 b^6 - 2747 b^4 - 1055 b^2 - 127 \right) b^6 a^{10} + 2 b^4 (b^{24} + 36 b^{22} \right. \\
& \left. + 180 b^{20} - 1372 b^{18} - 9036 b^{16} - 25400 b^{14} - 54798 b^{12} - 64184 b^{10} - 27212 b^8 - 6620 b^6 \right.
\end{aligned}$$

$$\begin{aligned}
& -2636 b^4 + 36 b^2 + 1) a^{12} - 4 b^2 (b^{28} + 12 b^{26} + 37 b^{24} + 378 b^{22} + 4187 b^{20} + 19656 b^{18} \\
& + 41514 b^{16} + 57358 b^{14} + 65514 b^{12} + 34120 b^{10} + 4315 b^8 + 1018 b^6 - 155 b^4 + 12 b^2 + 1) \\
& a^{14} - 4 b^2 (b^{28} + 76 b^{26} - 539 b^{24} - 2118 b^{22} + 5211 b^{20} + 36360 b^{18} + 92394 b^{16} + 78414 b^{14} \\
& + 38058 b^{12} + 17992 b^{10} - 2085 b^8 - 518 b^6 + 37 b^4 + 12 b^2 + 1) a^{18} + 8 \left(b^{20} + b^{18} - 699 b^{16} \right. \\
& + 2369 b^{14} - \frac{1037}{2} b^{12} - 1827 b^{10} + \frac{2163}{2} b^8 - 159 b^6 - 27 b^4 + b^2 + 1 \left. \right) b^6 a^{22} - 9 \left(b^{16} \right. \\
& + \frac{4168}{9} b^{14} - 528 b^{12} - 328 b^{10} + \frac{3698}{9} b^8 - 72 b^6 - 16 b^4 + 8 b^2 + 1 \left. \right) b^8 a^{24} + (-4 b^{18} \\
& + 8 b^{16} - 4 b^{14}) a^{30} + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} + (-1024 b^{22} + 72 b^{20} \\
& + 692 b^{18} + 8 b^{16} - 76 b^{14} + 72 b^{12}) a^{26} + (-254 b^{28} + 840 b^{26} + 6504 b^{24} - 696 b^{22} \\
& - 25752 b^{20} - 169584 b^{18} - 181788 b^{16} - 85104 b^{14} - 50328 b^{12} - 4792 b^{10} + 360 b^8 + 72 b^6 \\
& + 2 b^4) a^{20} + (b^{32} + 8 b^{30} + 20 b^{28} + 1672 b^{26} + 1950 b^{24} - 32376 b^{22} - 157780 b^{20} \\
& - 317080 b^{18} - 279672 b^{16} - 222104 b^{14} - 125524 b^{12} - 7288 b^{10} - 3426 b^8 - 120 b^6 + 20 b^4 \\
& + 8 b^2 + 1) a^{16} + (-9 b^{24} - 72 b^{22} + 144 b^{20} - 1912 b^{18} + 142 b^{16} + 4232 b^{14} + 2192 b^{12} \\
& - 1864 b^{10} - 265 b^8) a^8 + (72 b^{20} - 76 b^{18} + 1288 b^{16} + 692 b^{14} - 696 b^{12} - 512 b^{10}) a^6 \\
& + (6 b^{20} - 32 b^{18} + 44 b^{16} - 288 b^{14} - 250 b^{12}) a^4 + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^2 - 1024 (b^2 \\
& - 5) \left(b^2 - \frac{1}{2} \right) (b^2 + 1)^4 b^{13} a^{23} + 2560 \left(b^6 - \frac{6}{5} b^4 + \frac{8}{5} b^2 - 1 \right) (b^2 + 1)^4 b^7 a^{11} \\
& + 512 b^7 (b^2 - 2) (b^2 + 1)^5 a^9 + 512 b^7 (b^{10} - 10 b^8 - 9 b^6 + 42 b^4 - 19 b^2 + 1) (b^2 \\
& + 1)^4 a^{15} - 2560 (b^2 + 1)^4 \left(b^8 - \frac{16}{5} b^6 + b^4 - \frac{3}{5} b^2 + \frac{3}{5} \right) b^7 a^{13} + 512 b^7 (b^{10} + 11 b^8 \\
& - 78 b^6 + 81 b^4 - 22 b^2 + 1) (b^2 + 1)^4 a^{17} - 2560 \left(b^6 - \frac{38}{5} b^4 + \frac{42}{5} b^2 - \frac{11}{5} \right) (b^2 \\
& + 1)^4 b^{11} a^{21} - 1536 \left(b^8 - \frac{43}{3} b^6 + \frac{95}{3} b^4 - \frac{52}{3} b^2 + \frac{7}{3} \right) (b^2 + 1)^4 b^9 a^{19} \Big):
\end{aligned}$$

#Define $\tilde{B}$

$$\begin{aligned}
\tilde{B} := & \left(\left(-\frac{1}{2} b^5 + b^7 \right) a^{10} + \left(\frac{3}{2} b^7 - 3 b^5 + b^3 \right) a^8 + \frac{b^2 (b^2 - 5) (b^2 + 1)^3 a^7}{2} + \left(-\frac{1}{2} b \right. \right. \\
& \left. \left. - \frac{7}{2} b^5 + 3 b^3 \right) a^6 + \left(\frac{1}{2} - \frac{3}{2} b^8 - 4 b^6 - 3 b^4 \right) a^5 + \left(\frac{5}{2} b^3 - b - \frac{1}{2} b^7 \right) a^4 - (b^2 \right.
\end{aligned}$$

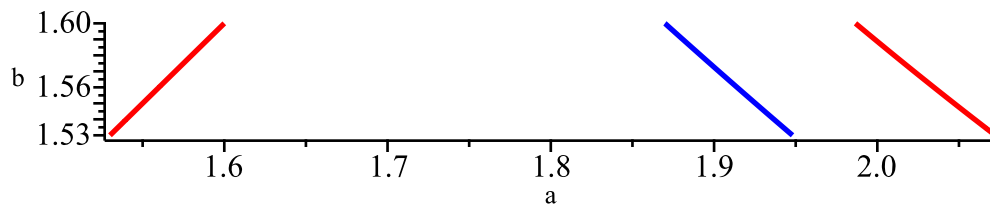
$$+ 1)^3 a^3 + \left(b^5 - \frac{1}{2} b \right) a^2 - \frac{b^3}{2} \Big) :$$

#Plot the curves $\tilde{n}=0$ and $\tilde{p}=0$ in region A5.

$r := a - b :$

with(plots) :

```
implicitplot([r2=0, B̃=0, r=0], a=1 .. 2.1, b=1.53 .. 1.6,
  scaling=constrained,
  gridrefine=2,
  thickness=2,
  color=[blue, red, red],
  labels=["a", "b"],
  title="Graphic for r2=0 and B̃=0 in A5");
Graphic for r2=0 and B̃=0 in A5
```



#Apply the Möbius transformation on the rectangle $A5 = [\sqrt{3}, 2.1) \times [1.53, 1.6)$ constructing $r2$

$$moebius := \begin{cases} a = \frac{\sqrt{3} + \left(\frac{21}{10}\right)c}{1 + c}, \end{cases}$$

$$b = \frac{\frac{153}{100} + \left(\frac{8}{5}\right) d}{1 + d} \Bigg\} :$$

```
r2 := collect(numer(expand(subs(moebius, r2))), c) :
```


[illegible]

$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$

$$seq(sign(evalf(coeff(c10, d, i))), i = 32 .. 0, -1);$$

[illegible]

$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$

$$seq(sign(evalf(coeff(c9, d, i))), i = 32 .. 0, -1);$$

[illegible]

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

$$seq(sign(evalf(coeff(c8, d, i))), i = 32 .. 0, -1);$$

[illegible]

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

$$seq(sign(evalf(coeff(c7, d, i))), i = 32 \dots 0, -1);$$

[illegible]

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

$$seq(sign(evalf(coeff(c6, d, i))), i = 32 .. 0, -1);$$

[illegible]

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

$$seq(sign(evalf(coeff(c5, d, i))), i = 32 .. 0, -1);$$

[illegible]

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

$$seq(sign(evalf(coeff(c4, d, i))), i = 32 .. 0, -1);$$

[illegible]

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

$$seq(sign(evalf(coeff(c3, d, i))), i = 32 \dots 0, -1);$$

[illegible]

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

$$seq(sign(evalf(coeff(c2, d, i))), i = 32 \dots 0, -1);$$

[illegible]

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

$$seq(sign(evalf(coeff(c1, d, i))), i = 32 \dots 0, -1);$$

[illegible]

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

$$seq(sign(evalf(coeff(c0, d, i))), i = 32 \dots 0, -1);$$

[illegible]

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

#Apply the Möbius transformation on the rectangle $[\sqrt{3}, 2) \times [1.53, 1.6)$ constructing B
 $B := \text{collect}(\text{numer}(\text{expand}(\text{subs}(\text{moebius}, \tilde{B}))), d) :$

#Compute the degree of B in c

$\text{degree}(B, c);$

10

(38)

#Define the coefficients $B_i = \text{coeff}(B, c, i)$
 $\text{seq}(\text{evalf}(\text{assign}(\text{cat}('B', i), \text{coeff}(B, c, i))), i = 0 .. 10);$

#Compute the all degree of B_i in d
 $\text{seq}(\text{degree}(\text{eval}(\text{cat}('B', i))), d), i = 0 .. 10);$

10, 10, 10, 10, 10, 10, 10, 10, 10, 10, 10

(39)

#From here we will analyze the signal variation of all coefficients B_i

$\text{seq}(\text{sign}(\text{coeff}(B10, d, i))), i = 10 .. 0, -1);$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(40)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B9, d, i))), i = 10 .. 0, -1);$

$\text{fsolve}(B9 = 0, d, 0 .. \text{infinity});$

1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1

0.1989786516

(41)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B8, d, i))), i = 10 .. 0, -1);$

$\text{fsolve}(B8 = 0, d, 0 .. \text{infinity});$

1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1

1.789056135

(42)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B7, d, i))), i = 10 .. 0, -1);$

-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1

(43)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B6, d, i))), i = 10 .. 0, -1);$

-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1

(44)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B5, d, i))), i = 10 .. 0, -1);$

-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1

(45)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B4, d, i))), i = 10 .. 0, -1);$

-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1

(46)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B3, d, i))), i = 10 .. 0, -1);$

-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1

(47)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B2, d, i))), i = 10 .. 0, -1);$

-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1

(48)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B1, d, i))), i = 10 .. 0, -1);$

-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1

(49)

$$seq(sign(evalf(coeff(B0, d, i))), i = 10 .. 0, -1);$$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

(50)